



Vanguard Knowledge Management

Vendor Evaluation Criteria

For vendor review

Date: June 25, 2026

Classification: Confidential —for evaluation participants only

1. Purpose

This document covers the Set-Based Design for the POC (Proof of Concept) that vendors will follow for Project Vanguard's Knowledge Management Capability. The POC will consist of two evaluation stages that evaluate the vendors' ability to convert large volumes of open-source data into a queryable knowledge graph and answer tiered-difficulty questions over that graph using graph-based retrieval-augmented generation (GraphRAG) and agentic, multi-step reasoning.

The purpose of this document is to give participating vendors a single, unambiguous statement of what the POC is testing, how the evaluation is staged, what artifacts vendors must submit, the templates those artifacts must follow, and the timeline for response. It is written so that vendors are scored on the same criteria, on representative data, with results that Cotiviti can audit and compare side by side.

2. How the POC works

The POC runs in two evaluation stages. Stage 1 uses a publicly available, open-source sample. Stage 2 uses Cotiviti-generated synthetic medical records. Vendors must pass Stage 1 to receive Stage 2 data, and vendors that clear both will have officially completed the POC and enter the prototyping phase (Section 5). This section sets out what every vendor will be evaluated on, the evaluation domains and the gating metrics that apply across the POC. Sections 3 and 4 will detail exactly what each stage requires.

2.1 Evaluation domains

Vendors are evaluated across four domains: two functional (graph construction, and retrieval and reasoning), the non-functional requirements (ingestion throughput, latency, and cost), and safety, which is cross-cutting and gating—it must hold across every layer and cannot be lost in any one of them. Each domain is scored on its own; a single composite score is not used.

Domain	Layer	What it covers
Graph construction	Functional	Entity and relation extraction, concept normalization, clinical context accuracy, coreference and entity resolution, schema conformance, logical consistency, coverage, connectivity, and provenance completeness.
Retrieval and reasoning	Functional	Retrieval quality against the correct subgraph, multi-hop and temporal reasoning, answer faithfulness and correctness, citation accuracy, and agentic and conversational robustness.
Reported non-functional requirements	Non-functional	Ingestion throughput, query latency (p50, p95, p99), incremental update without full

Commented [HK1]: says non-functional here under 'domain'

but layer in the others says 'functional'

Commented [HK2R1]: perhaps add another col 'Functional' ?

Commented [ZB3R1]: Not sure if I understand

Commented [ZB4R1]: Made the Layer column consistent —graph construction and retrieval/reasoning are Functional, the reported requirements are Non-functional, and safety is Cross-cutting (gating). No extra column needed.

Domain	Layer	What it covers
		re-ingestion, cost per document, and graph-store behavior at steady-state node and edge counts.
Safety	Cross-cutting (gating)	Clinical context accuracy as a safety metric, provenance auditability, and resistance to prompt injection in the agentic layer.

3. Stage 1: Open-source data sample

This stage consists of: Cotiviti will issue two publicly available, open-source datasets, together with the submission templates in Section 8. Stage 1 has two tracks that exercise different capabilities, and vendors must complete both.

Track A —Query a provided knowledge graph (PrimeKG): Cotiviti provides PrimeKG, a pre-built biomedical knowledge graph. Vendors do not build a graph here; they load it and answer questions by querying it. This track tests retrieval and reasoning over an existing graph.

Track B —Build a knowledge graph (MultiHop RAG): Vendors build their own knowledge graph from the MultiHop RAG open-source dataset, then answer multi-hop questions over the graph they constructed. This track tests both graph construction and multi-hop retrieval and reasoning.

What the vendor will submit: For Track A, answers to the 100-question set and the reasoning traces behind them —no constructed graph is required, since the graph is provided. For Track B, the constructed knowledge graph, answers to the 57-question multi-hop set, and the reasoning traces behind those answers. Both tracks include a self-reported metrics and operations summary. The full package and required formats are set out below.

Evaluation: The Stage 1 gating metrics are provenance completeness and schema conformance (Track B graph), and citation accuracy on answers (both tracks). Graph-construction metrics apply to Track B only. You must pass Stage 1 to receive Stage 2 data.

Submission package. Vendors submit the following package at each stage; formats are defined in the submission templates (Section 8).

Artifact	Format	Stage(s)
Constructed knowledge graph (nodes, edges, manifest)	JSON Lines + manifest (Section 8.1)	1 (Track B only) and 2
Question-and-answer results	JSON Lines (Section 8.2)	1 and 2
Reasoning traces	JSON (Section 8.3)	1 and 2
Metrics self-report	Completed table (Section 8.4)	1 and 2

Artifact	Format	Stage(s)
Operations report (throughput, latency p50/p95/p99, cost per document, hardware, and configuration)	PDF or Markdown	1 and 2
Methodology summary (models, ontologies and versions, pipeline architecture, and how external document text is isolated from agent instructions)	PDF or Markdown	1 and 2
Clarification log (questions raised and Cotiviti responses relied upon)	PDF or Markdown	1 and 2

Graph-construction metrics. These metrics apply to Track B (the MultiHop RAG graph vendors build) only; they do not apply to Track A, where the graph (PrimeKG) is provided. The table below lists the full set of graph-construction dimensions Cotiviti cares about in general. Note that for this POC no gold reference graph is supplied for MultiHop RAG, so the accuracy-based rows that require a reference (entity-extraction F1, concept normalization / entity linking, relation-extraction F1, coreference / entity resolution, logical consistency) cannot be scored and are shown for reference only. The rows actually evaluated on the Track B graph are the structural and descriptive ones—schema conformance, provenance completeness, coverage, and connectivity—which Cotiviti verifies directly from the submitted files. Of these, provenance completeness and schema conformance are gating. Clinical context accuracy does not apply, since the MultiHop RAG data is non-clinical.

Layer	Metrics
Entity extraction	Precision, recall, F1 per entity type
Concept normalization / entity linking	Accuracy at rank one and rank k against the correct clinical coding; NIL handling
Relation extraction	Precision, recall, F1 per relation type
Clinical context accuracy	Negation, uncertainty, temporality, experiencer
Coreference / entity resolution	False merges and false splits, across the full document
Schema conformance	Ontology and constraint-violation rate
Logical consistency	Contradiction checks, ideally via an OWL reasoner
Coverage / completeness	Fraction of source facts represented
Connectivity	Orphan-node rate, cross-page linkage
Provenance completeness	Every node and edge traceable to page and span

4. Stage 2: Synthetic medical records

This stage consists of: Vendors that pass Stage 1 receive Cotiviti-generated synthetic medical records that mirror our production distribution.

Data distribution: Stage 2 records are stratified across specialties, document lengths, and complexity, and deliberately include full multi-thousand-page journeys to stress entity resolution and long-context linkage. The distribution reflects production, including messy and degraded documents; clean records are not oversampled.

The questions. Stage 2 exercises the full question taxonomy below; vendors are scored per category rather than on a single blended number. *(Stage 1 uses two open-source datasets: Track A queries the provided PrimeKG graph over 100 questions of increasing complexity, and Track B builds a graph from MultiHop RAG and answers 57 multi-hop questions.)*

Category	What it assesses	Why it matters
Single-hop factoid	Direct lookup of one fact	Baseline retrieval
Multi-hop traversal	Reasoning across linked nodes	Core GraphRAG capability
Temporal / aggregation	Trends across an admission	e.g., hemoglobin trend over time
Cross-document	Facts spanning multiple records	Patient-journey linkage
Comparison	Relating two or more entities	Synthesis
Absence / negation	Confirming something is absent	Where clinical retrieval silently fails

What the vendor will submit: The same package as Stage 1, now runs against the synthetic records.

Evaluation: The Stage 2 gating metrics are clinical context accuracy, provenance completeness, and the prompt-injection slice (Section 2.2), scored against Cotiviti’s retained ground truth.

Retrieval and reasoning metrics. Stage 2 answers will be scored on the criteria below.

Layer	Metrics	Notes
Retrieval	Precision at k, recall at k, mean reciprocal rank (MRR), normalized discounted cumulative gain (nDCG)	Against the correct subgraph
Multi-hop reasoning	Accuracy on the traversal question slice	Core GraphRAG capability
Answer quality	Faithfulness / groundedness, answer correctness, context precision and recall, citation accuracy	Citations must resolve to source
Agentic and multi-step	Task-completion rate; trajectory (step-level) correctness	A right answer via an invalid path will not generalize

Layer	Metrics	Notes
Conversational robustness	Context retention across turns; consistency across paraphrases	Multi-turn scenarios

5. After the POC: Prototype

Vendors that pass both Stage 1 and Stage 2 will be submitted to Cotiviti’s third-party (3P) review and procurement process. Once a vendor clears 3P review, they advance into the next stage of the set-based design: building a prototype, and testing with real PHI. The prototype phase will be governed by a separate document.

Commented [HK5]: will a new document be created for prototype phase?

Commented [ZB6R5]: Yes

Commented [ZB7R5]: Added a line to Section 5: the prototype phase will be governed by a separate document.

6. Performance and safety requirements

The functional scoring rubrics sit with the stages they apply to graph-construction metrics under Stage 1 (Section 3) and retrieval-and-reasoning metrics under Stage 2 (Section 4). This section covers the two requirement sets that apply across both stages, which are reported non-functional requirements and safety. Cotiviti retains the ground truth and scores vendor answers against it; a single composite score is not used.

6.1 Reported non-functional requirements

Cotiviti’s volume is the real constraint, so operational metrics carry real weight.

Layer	Metrics	Notes
Ingestion throughput	Pages per second; sustained documents per day	At target volume
Query latency	p50, p95, p99	Report all three
Incremental update	Behavior on amended records	Must not require full re-ingestion
Cost and scalability	Cost per document; graph-store behavior at steady-state node and edge counts	Report configuration and hardware assumptions

6.2 Safety

Ingested clinical documents are untrusted external input and will be treated as potentially malicious. Two safety items are built into the benchmark, in addition to the clinical context accuracy metric above.

- **Prompt-injection slice (gating).** Documents containing text crafted to manipulate a downstream agent, for example, instructions embedded in a free-text note. The agentic layer must not treat graph or document content as trusted instructions.

- **Clinical context accuracy (gating).** Carried from the graph-construction metrics in Section 3; a negated or family-history finding rendered as a positive patient fact is a safety failure.

7. Submission and Reporting Requirements

7.1 Format and delivery conventions

- **Machine-readable first.** Graph, Q&A, and reasoning-trace artifacts need to be submitted as JSON or JSON Lines (one record per line) using the schemas in Section 8. Identifiers must be stable and cross-referenced across files.
- **File naming.** Use vendor_<vendorid>_<stage>_<artifact>_v<n>.<ext>, for example vendor_acme_stage1_graph-nodes_v1.jsonl.
- **Encoding.** UTF-8, Unix line endings, one JSON object per line for JSON Lines files.
- **Versioning.** Increment the version suffix on resubmission; do not overwrite. State which version is authoritative.
- **Delivery.** Submit through the secure upload channel provided at kickoff. Do not send artifacts by email or unsecured links.
- **Completeness.** Partial submissions will be scored as submitted; missing gating artifacts (graph provenance, reasoning traces) are treated as a gate failure.

8. Submission Templates

The following schemas are for reference only and are not strict formats. Identifiers (doc_id, node_id, edge_id, question_id, trace_id) must be consistent across all files so that an answer can be traced to a reasoning path, to the graph nodes and edges it used, and to the source page and span. Fields shown are the minimum required; vendors may add fields but must not rename or omit required ones.

Which templates apply where. Stage 1 Track A (PrimeKG) submits only the Q&A results (8.2) and reasoning traces (8.3); the graph is provided, so no graph submission (8.1) or graph-construction metrics are required. Stage 1 Track B (MultiHop RAG) and Stage 2 submit the full set, including the constructed graph (8.1). The clinical-context block in 8.1/8.3 applies only to clinical data (PrimeKG in Track A is clinical/biomedical, and Stage 2 records are clinical); for the non-clinical MultiHop RAG data in Track B, omit clinical-context fields. For graph-native sources such as PrimeKG that have no page or character span, cite the source node, edge, or record identifier instead (see the provenance note in 8.1).

8.1 Graph submission format

The graph needs to be submitted as two JSON Lines files (nodes and edges) plus a manifest. Every node and edge carries provenance —this is the gating provenance-completeness requirement. This applies to Stage 1 Track B (MultiHop RAG) and Stage 2;

Stage 1 Track A does not submit a graph, since PrimeKG is provided. Every node and edge must carry a provenance pointer to the source document it came from. For the MultiHop RAG articles, the document identifier (doc_id) is required and the character span within the article is optional but encouraged, since the dataset's evidence is defined at the article level. For graph-native sources such as PrimeKG, cite the source node, edge, or record identifier (a source_ref) instead of page and span. Every node and edge must carry at least one such pointer. The clinical_context block is required only for clinical data and should be omitted for the non-clinical MultiHop RAG data.

Nodes —one JSON object per line (graph-nodes.jsonl)

```
{
  "node_id": "n_condition_902",
  "type": "Condition",
  "label": "Sepsis",
  "normalized_code": { "system": "SNOMED CT", "code": "91302008", "display": "Sepsis" },
  "nil": false,
  "clinical_context": {
    "negation": false,
    "uncertainty": "confirmed",
    "temporality": "current",
    "experiencer": "patient"
  },
  "provenance": { "doc_id": "DOC-0003", "page": 5, "span": [210, 216] }
}
```

For the non-clinical data, the same shape is used but the fields carry news-domain values: for example a node with type "Organization" and label "Spotify", no normalized_code and no clinical_context, and provenance {"doc_id": "article_00123"} pointing to the source article; an edge such as {"predicate": "reported_by", "subject": "n_org_spotify", "object": "n_source_techcrunch", "provenance": {"doc_id": "article_00123"}}.

Edges —one JSON object per line (graph-edges.jsonl)

```
{
  "edge_id": "e_hasDiagnosis_77",
  "subject": "n_encounter_551",
  "predicate": "encounter_has_diagnosis",
  "object": "n_condition_902",
  "provenance": { "doc_id": "DOC-0003", "page": 5, "span": [180, 240] }
}
```

Manifest —single JSON object (graph-manifest.json)

```
{
  "vendor_id": "acme",
  "stage": 1,
  "schema_version": "graph-1.0",
  "dataset": "multihop_rag", "source_vocabularies": { "note": "list vocabularies/ontologies used by the source and their versions, if any; e.g. SNOMED CT / RxNorm / LOINC for clinical data" },
  "counts": { "documents": 50, "nodes": 18432, "edges": 27110, "nil_mentions": 214 },
  "notes": "free-text caveats, if any"
}
```

8.2 Question-and-answer results template

One JSON object per line, one per question in the provided question set. The answer must reference the reasoning trace that produced it and the graph elements it used. The

example below is clinical and illustrative; the same structure is used for every stage and track. For Track A (PrimeKG), citations and retrieved_context point to PrimeKG node/edge/record ids via source_ref rather than page and span; for the non-clinical Track B data, the clinical_context and context_check fields are omitted.

Q&A results—one JSON object per line (qa-results.jsonl)

```
{
  "question_id": "Q-MH-0042",
  "category": "multi_hop_traversal",
  "question": "Was there any evidence of sepsis during the April 2023 admission?",
  "vendor_answer": "Yes – sepsis was documented on the 2023-04-11 admission.",
  "answer_type": "boolean_with_evidence",
  "confidence": 0.91,
  "retrieved_context": [
    { "doc_id": "DOC-0003", "page": 5, "span": [180, 240], "snippet": "..."}
  ],
  "citations": [ { "doc_id": "DOC-0003", "page": 5, "span": [210, 216] } ],
  "graph_nodes_used": [ "n_encounter_551", "n_condition_902" ],
  "graph_edges_used": [ "e_hasDiagnosis_77" ],
  "reasoning_trace_ref": "T-MH-0042",
  "latency_ms": 820
}
```

8.3 Reasoning-trace template

One JSON object per answer. Each trace is an ordered list of steps; each step records the operation, the graph element involved, and the evidence. Trajectories are graded step by step, so a correct answer reached through an invalid path is detectable.

Reasoning trace (reasoning-traces.json —array, or one object per line)

```
{
  "trace_id": "T-MH-0042",
  "question_id": "Q-MH-0042",
  "steps": [
    {
      "step": 1,
      "operation": "entity_lookup",
      "input": "April 2023 admission",
      "node_id": "n_encounter_551",
      "node_type": "Encounter",
      "evidence": { "doc_id": "DOC-0003", "page": 2, "span": [88, 140] }
    },
    {
      "step": 2,
      "operation": "edge_traversal",
      "edge_id": "e_hasDiagnosis_77",
      "predicate": "encounter_has_diagnosis",
      "from": "n_encounter_551",
      "to": "n_condition_902"
    },
    {
      "step": 3,
      "operation": "context_check",
      "node_id": "n_condition_902",
      "clinical_context": { "negation": false, "experiencer": "patient", "temporality": "current" }
    }
  ],
  "final_answer": "Yes – sepsis documented on the 2023-04-11 admission.",
}
```

```
"answer_supported_by": ["e_hasDiagnosis_77", "n_condition_902"]
}
```

8.4 Metrics self-report template

This table is a vendor self-report: complete only the rows a vendor can actually measure from its own system and submitted files, for the stage and track being submitted. Cotiviti scores answer correctness, multi-hop accuracy, and citation quality separately, against a withheld answer key; the "Answer and citation quality" row is listed only so vendors know what their answers are judged on, and is not self-reported. Because no gold reference graph is provided, graph-construction accuracy metrics (entity-extraction F1, entity-linking accuracy, relation-extraction F1) are not reported for the self-built Track B graph — there is nothing to score them against. The graph-construction rows that remain (schema conformance, provenance completeness, coverage/connectivity) are structural or descriptive checks vendors can run on their own files without an answer key. Non-functional metrics (throughput, latency, cost) must be reported together with the hardware, model, and concurrency that produced them, since these numbers depend on the machine used and are otherwise not comparable. The "Applies to" column shows where each row is required —"Track A" and "Track B" are the two Stage 1 tracks, and "All" means every stage and track. Track A (PrimeKG, provided graph) has no graph-construction or ingestion rows, since the graph is provided.

Domain	Metric	How to compute / what to report	Applies to
Graph construction	Schema conformance (gating)	A structural check on your own submitted files —no answer key needed. Report the rate at which nodes/edges violate the required schema: unknown types, missing required fields, dangling edge endpoints, malformed identifiers. Report violations per 1,000 elements; lower is better.	Track B; Stage 2
Graph construction	Provenance completeness (gating)	A structural check on your own files —no answer key needed. Report the share of nodes and edges that carry a resolvable provenance pointer back to a source document (doc_id). Report as a percentage; target is 100%. Any node or edge with no doc_id counts as incomplete.	Track B; Stage 2
Graph construction	Coverage and connectivity	Self-reported descriptive stats on your graph: node and edge counts by type, orphan-node rate (nodes with no edges),	Track B; Stage 2

Domain	Metric	How to compute / what to report	Applies to
		and the share of source documents represented by at least one node. Descriptive only, not scored against a key.	
Retrieval and reasoning	Answer and citation quality	Scored by Cotiviti against the withheld answer key —do not self-report a value. Listed here so you know what your answers are judged on: answer correctness, multi-hop accuracy, and whether your citations resolve to the correct source article(s).	All (Cotiviti scores)
Reported NFRs	Ingestion throughput (with config)	Self-reported. Pages or documents per unit time for building the graph, reported together with the hardware and settings that produced it: machine type / vCPU / GPU / RAM, model(s) used, and concurrency. A number without its configuration is not comparable.	Track B; Stage 2
Reported NFRs	Query latency p50 / p95 / p99 (with config)	Self-reported. Wall-clock time from question in to answer out, over the full question set, as p50/p95/p99 in milliseconds. Report the same hardware/model/concurrency context, since latency depends on the machine used. State the number of queries measured.	All
Reported NFRs	Cost (with config)	Self-reported. Fully-loaded cost per document ingested and, where relevant, per query —model/API + compute + storage. State the pricing basis and the configuration assumed. Report the estimate; exact figures are not scored, but the basis must be stated.	All

9. Timeline and Milestones

Timing will be in business days from the Stage 1 kickoff (Day 0); exact calendar dates are confirmed in the kickoff notice.

Phase / milestone	Timing	Vendor action
Stage 1 kickoff: POC Eval, open-source sample, and templates released	Day 0	Confirm receipt
Stage 1 submission deadline	Day 5 (end of day, ET)	Submit full Stage 1 package (Section 3)
Cotiviti Stage 1 evaluation	Day 6 to Day 8	None
Stage 1 results and Stage 2 invitations	Day 9	Selected vendors confirm receipt
Stage 2 kickoff: synthetic records released	Day 10	Begin Stage 2 run
Stage 2 submission deadline	Day 15 (End of Day, ET)	Submit full Stage 2 package
Cotiviti Stage 2 evaluation	Day 25 to Day 31	None
Stage 2 results; advance to 3P review	Day 32	Selected vendors enter Cotiviti 3P process
Third-party review and prototype scoping	Day 33 onward	Per 3P process